

수소경제와 에너지저장 시스템

(Hydrogen Economy and Energy Storage Systems)

Haewon Seo, Ph. D

Hydrogen **E**nerg**Y** Lab **ON** **E**arth

Assistant Professor

Division of Climate Change

Hankuk University of Foreign Studies (HUFS)

Course Introduction

❑ Lecture: Hydrogen Economy and Energy Storage Systems (L01212101)

- Classroom: 0413
- Class time: Tuesday, 12:00–14:50

❑ Professor: Haewon Seo

- Office: Prof Res Bldg , Rm 202
- E-mail: heyone@hufs.ac.kr
- Office hour: Monday (till 15:00)



❑ Scope

- This course introduces hydrogen energy and its associated devices for a clean and sustainable future. Based on fundamental electrochemistry, it examines the operating principles of energy conversion systems for hydrogen production and utilization. Furthermore, it reviews the latest technological trends and explores the development of next-generation energy conversion technologies.

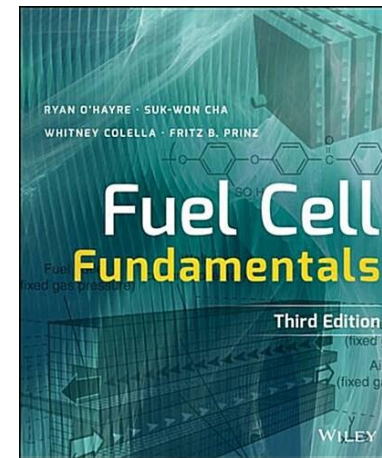
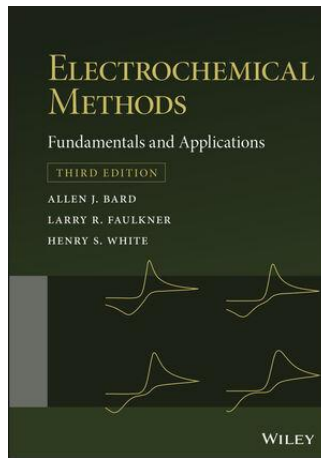
Course Introduction



□ Textbook

– Main

- Electrochemical Methods (A.J. Bard et al.), 3rd Ed., Wiley (2022)
- Thermodynamics (Y.A. Cengel et al.), 9th Ed., McGraw-Hill (2018)
- Fuel Cell Fundamentals (R. O'Hayer et al.), 3rd Ed., Wiley (2016)



– Reference

- Scientific articles
- IRENA, IEA, World Energy Council, DOE, etc.

Course Introduction

□ **Schedule**; to be flexibly operated

Week	Date	Contents
01st	Sep 01st 2026	Introduction to Hydrogen (1)
02 nd	Sep 08 th 2026	Introduction to Hydrogen (2)
03 rd	Sep 15 th 2026	Introduction to Hydrogen (3)
04 th	Sep 22 nd 2026	Fundamentals of Electrochemistry
05 th	Sep 29 th 2026	Thermodynamics of Reacting Systems
06 th	Oct 06 th 2026	Electrochemical Cells & Potentials
07 th	Oct 13 th 2026	Reference Electrodes & Practical Cells
08 th	Oct 20 th 2026	Midterm exam
09 th	Oct 27 th 2026	Cell Polarization & Overpotentials
10 th	Nov 03 rd 2026	Electrode Kinetics & Interface (EDL)
11 th	Nov 10 th 2026	Mass Transport & Chronoamperometry
12 th	Nov 17 th 2026	Voltammetry & Reaction Mechanisms
13 th	Nov 24 th 2026	Electrochemical Impedance
14 th	Dec 01 st 2026	Applications
15 th	Dec 15 th 2026	Final exam

Course Introduction

Evaluation

- Attendance (10%)
- Assignment (20%)
- Midterm Exam (30%)
- Final Exam (40%)

1. Introduction to Hydrogen (1)

Contents

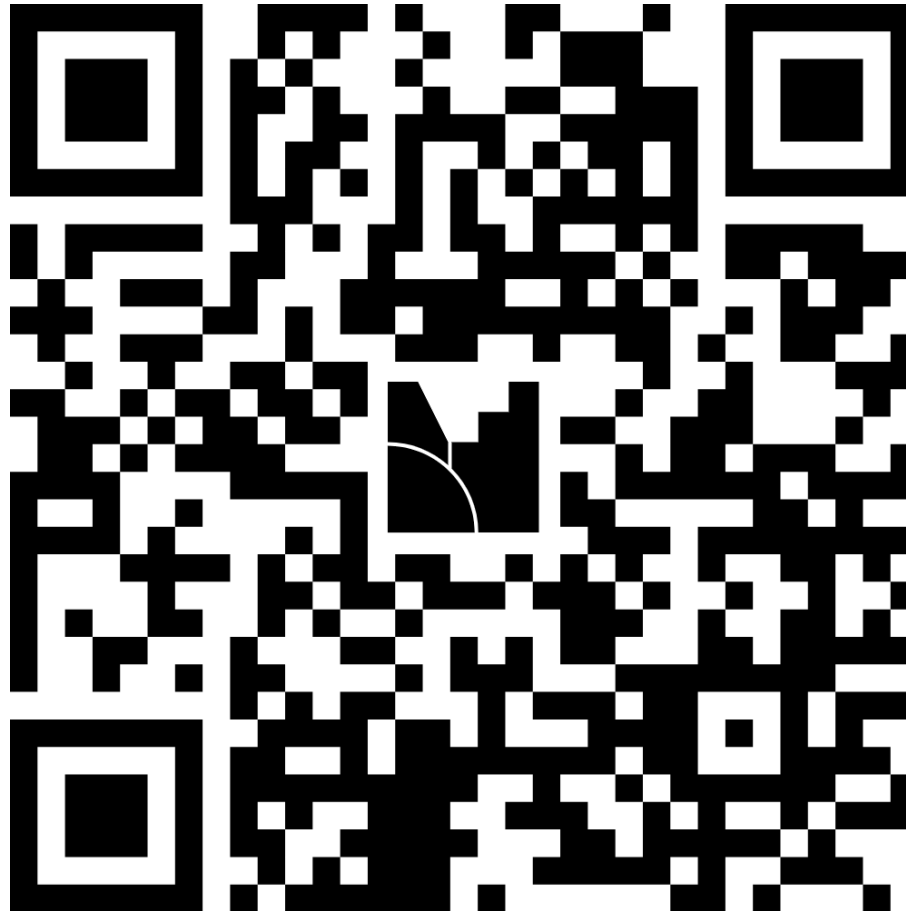


1. Overview of Hydrogen

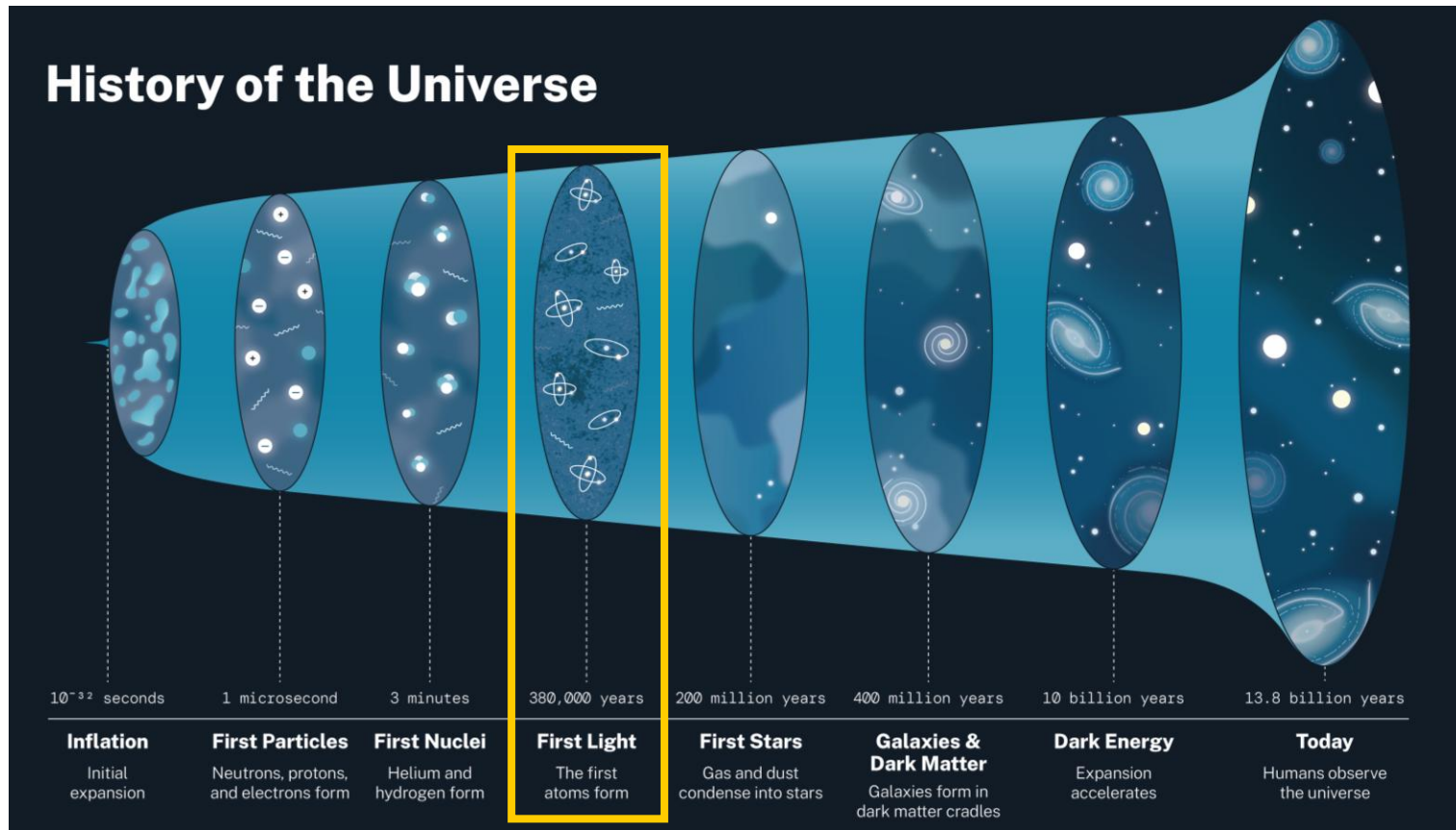
Overview of Hydrogen



<https://www.mentimeter.com/app/presentation/alffegrpwxtrmxwbokrtq7n7fe4et78v2/present?question=9oz5vx96zvy6>

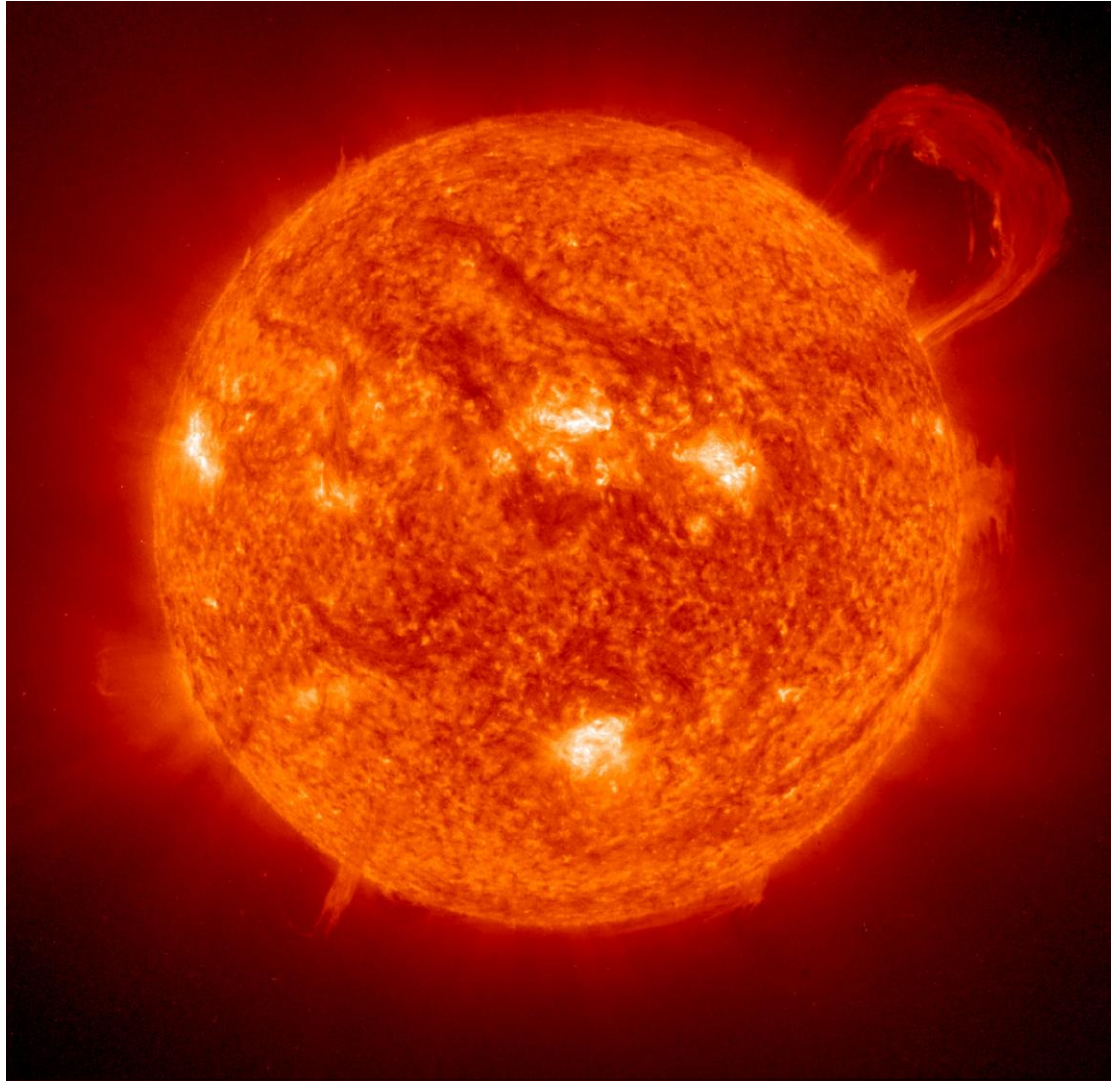


Overview of Hydrogen



- 10^{-35} – 10^{-5} s (Hadron epoch): Formation of protons & neutrons (Quark coupling)
- 10^{-5} – 10^0 s (Lepton epoch): Temperature drop due to further expansion
- 10^0 – 10^2 s: Formation of atomic nuclei (Nucleosynthesis)
- 10^2 s–380,000 yr: Mass ratio shifts (Proton/Neutron = 14:2 → H/He = 75:25 wt%)
- **After 380,000 yr: Temperature drop and Formation of atom**

Overview of Hydrogen



<Source: NASA>

Overview of Hydrogen

□ Hydrogen

The simplest element in the universe (one proton, one electron, and no neutron)

periodic table of the elements

Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 수소 hydrogen																	2 He 헬륨 helium
2	3 Li 리튬 lithium	4 Be 베릴륨 beryllium																10 Ne 네온 neon
3	11 Na 나트륨 sodium	12 Mg 마그네슘 magnesium																18 Ar 아르곤 argon
4	19 K 칼륨 potassium	20 Ca 칼슘 calcium	21 Sc 스칸듐 scandium	22 Ti 타이타늄/티탄 titanium	23 V 바나듐 vanadium	24 Cr 크로뮴/크롬 chromium	25 Mn 망가니즈/망간 manganese	26 Fe 철 iron	27 Co 코발트 cobalt	28 Ni 니켈 nickel	29 Cu 구리 copper	30 Zn 아연 zinc	31 Ga 갈륨 gallium	32 Ge 저마늄/게르마늄 germanium	33 As 비소 arsenic	34 Se 셀레늄/셀렌 selenium	35 Br 브로민/브롬 bromine	36 Kr 크립톤 krypton
5	37 Rb 루비듐 rubidium	38 Sr 스트론튬 strontium	39 Y 이트륨 yttrium	40 Zr 지르코늄 zirconium	41 Nb 나이오븀 niobium	42 Mo 몰리브덴 molybdenum	43 Tc 테크네튬 technetium	44 Ru 루테튬 ruthenium	45 Rh 로듐 rhodium	46 Pd 팔라듐 palladium	47 Ag 은 silver	48 Cd 카드뮴 cadmium	49 In 인듐 indium	50 Sn 주석 tin	51 Sb 안티모니/안티몬 antimony	52 Te 텔루륨/텔레루 tellurium	53 I 아이오딘/요오드 iodine	54 Xe 제논/크세논 xenon
6	55 Cs 세슘 caesium	56 Ba 바륨 barium	57-71 lanthanoids	72 Hf 하프늄 hafnium	73 Ta 탄탈럼/탄탈 tantalum	74 W 텅스텐 tungsten	75 Re 레늄 rhenium	76 Os 오스뮴 osmium	77 Ir 아일랜드 iridium	78 Pt 백금 platinum	79 Au 금 gold	80 Hg 수은 mercury	81 Tl 탈륨 thallium	82 Pb 납 lead	83 Bi 비스무트 bismuth	84 Po 폴로늄 polonium	85 At 아스타틴 astatine	86 Rn 라돈 radon
7	87 Fr 프랑슘 francium	88 Ra 라듐 radium	89-103 actinoids	104 Rf 러더포듐 rutherfordium	105 Db 두브늄 dubnium	106 Sg 시보그뮴 seaborgium	107 Bh 보륨 bohrium	108 Hs 하슘 hassium	109 Mt 마이트너뮴 meitnerium	110 Ds 다름슈타뮴 darmstadtium	111 Rg 뢴트게늄 roentgenium	112 Cn 코페르니슘 copernicium	113 Nh 니호늄 nihonium	114 Fl 플레로븀 flerovium	115 Mc 모스크로븀 moscovium	116 Lv 리버모븀 livermorium	117 Ts 테네시 tennessine	118 Og 오가네손 oganesson
				57 La 란타넘 lanthanum	58 Ce 세륨 cerium	59 Pr 프라세오디뮴 praseodymium	60 Nd 네오디뮴 neodymium	61 Pm 프로메튬 promethium	62 Sm 사마륨 samarium	63 Eu 유로뮴 europium	64 Gd 가돌리늄 gadolinium	65 Tb 터븀/테르븀 terbium	66 Dy 디스프로슘 dysprosium	67 Ho 홀뮴 holmium	68 Er 에르븀/에르븀 erbium	69 Tm 툴륨 thulium	70 Yb 이트륨/이트레븀 ytterbium	71 Lu 루테튬 lutetium
				89 Ac 악티늄 actinium	90 Th 토륨 thorium	91 Pa 프로트악티늄 protactinium	92 U 우라늄 uranium	93 Np 넵투늄 neptunium	94 Pu 플루토늄 plutonium	95 Am 아메리슘 americium	96 Cm 퀴륨 curium	97 Bk 버클륨 berkelium	98 Cf 캘리포늄 californium	99 Es 아인슈타이늄 einsteinium	100 Fm 페르뮴 fermium	101 Md 멘델레븀 mendelevium	102 No 노벨륨 nobelium	103 Lr 로렌슘 lawrencium

Legend:

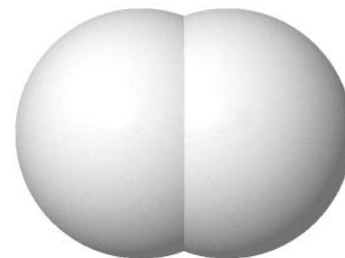
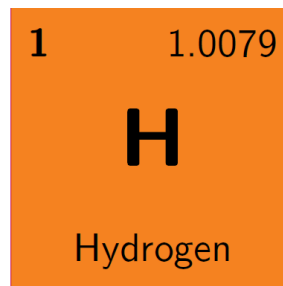
- 기체 (Gas): H, He, Ne, Ar, Kr, Xe, Rn
- 액체 (Liquid): Hg
- 고체 (Solid): All other elements
- 준금속 (Metalloid): B, Si, As, Sb, Te, Po
- 비금속 (Non-metal): C, N, O, F, Cl, Br, I, At
- 알칼리금속 (Alkali metal): Li, Na, K, Rb, Cs, Fr
- 알칼리토금속 (Alkaline earth metal): Be, Mg, Ca, Sr, Ba, Ra
- 전이금속 (Transition metal): Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, Po, At, Rn, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr
- 전이후금속 (Post-transition metal): Al, Si, P, S, Cl, Ar, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, Po, At, Rn, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr
- 비활성기체 (Noble gas): He, Ne, Ar, Kr, Xe, Rn

Overview of Hydrogen

□ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: $-259.2\text{ }^{\circ}\text{C}$
- Boiling Point: $-252.9\text{ }^{\circ}\text{C}$
- Density: 0.090 g L^{-1} (at STP ($0\text{ }^{\circ}\text{C}$, 1 atm))



–Typically, hydrogen refers to *hydrogen gas* (H_2), which is a colorless, odorless, and nontoxic gas

Table Physical properties of gases at standard temperature and pressure (STP) ($0\text{ }^{\circ}\text{C}$, 1 atm)

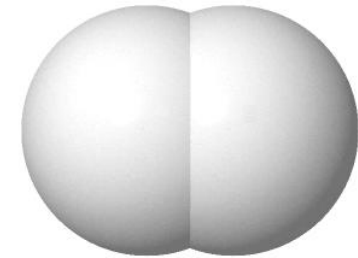
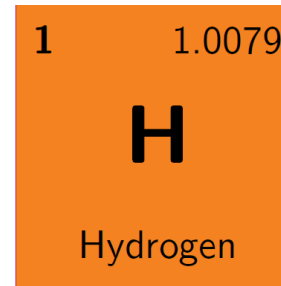
Gas	Density, ρ (g L^{-1})	Specific heat capacity, c_v ($\text{J g}^{-1}\text{ K}^{-1}$)	Viscosity, μ ($\mu\text{Pa s}$)	Thermal cond., k ($\text{W m}^{-1}\text{ K}^{-1}$)
H_2	0.090	10.16	8.4	0.168
He	0.179	3.11	18.6	0.142
CH_4	0.717	1.70	10.3	0.030
Air	1.292	0.718	17.2	0.024
O_2	1.429	0.655	19.2	0.024
CO_2	1.977	0.634	13.7	0.015

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Density

$$\rho \propto m$$

Specific heat capacity

$$c_v \propto \frac{1}{m}$$

Viscosity (Momentum)

$$\mu = \alpha \rho \lambda \sqrt{\frac{2k_B T}{\pi m}} \propto \frac{\rho \lambda}{\sqrt{m}} = \frac{\sqrt{m}}{d^2}$$

Thermal conductivity (Energy)

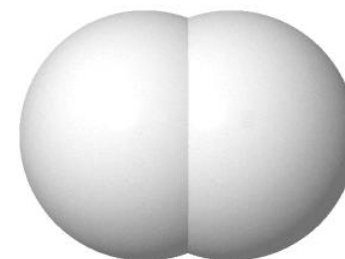
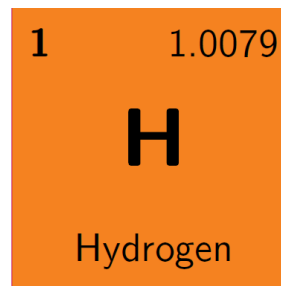
$$k = \beta \rho \lambda c_v \sqrt{\frac{2k_B T}{\pi m}} \propto \frac{\rho \lambda c_v}{\sqrt{m}} = \frac{1}{\sqrt{m} d^2}$$

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Table Physical properties of liquids

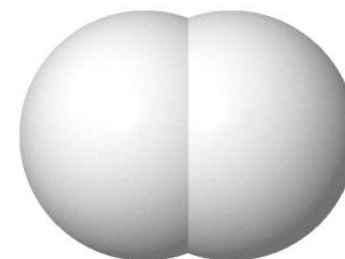
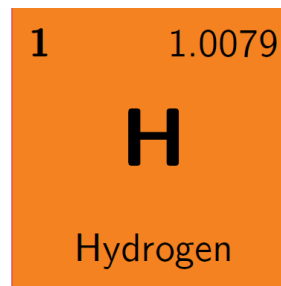
Liquid	Density, ρ (g L^{-1})	Boiling point ($^{\circ}\text{C}$)	Remarks
H_2	70.85	-252.9	$\sim 1/6$ of CH_4's density
He	125.0	-268.9	
CH_4	422.8	-161.5	LNG
Air	874.0	-194.3	
O_2	1141	-183.0	
CO_2	1178	-78.5	At triple-point

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H_2 makes up ~0.5–0.6 ppm of the Earth's atmosphere, whereas H is ubiquitous in water (H_2O), organic compounds (CHONPS), and other matter.

Table Major constituents of the atmosphere of Earth in parts-per million (ppm)

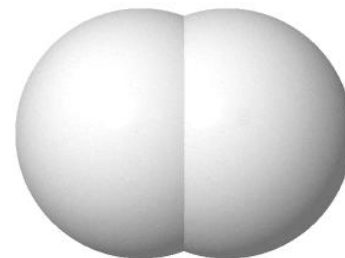
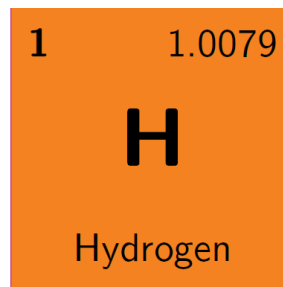
Gas	Volume fraction (ppm)	Gas	Volume fraction (ppm)	Gas	Volume fraction (ppm)
N_2	780,840	CH_4	1.92	O_3	0.07
O_2	209,460	Kr	1.14	NO_2	0.02
Ar	9,340	H_2	0.55	H_2O	Up to 40,000
CO_2	420	N_2O	0.33		
Ne	18.182	CO	0.10		
He	5.24	Xe	0.09		

Overview of Hydrogen

☐ Hydrogen

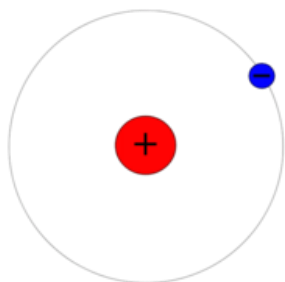
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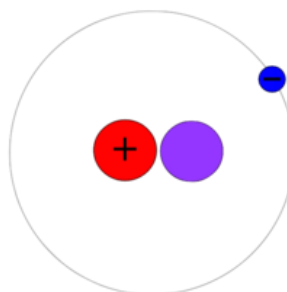


–Hydrogen has three main naturally occurring isotopes: ^1H (protium), ^2H (D) (deuterium), ^3H (T) (tritium)

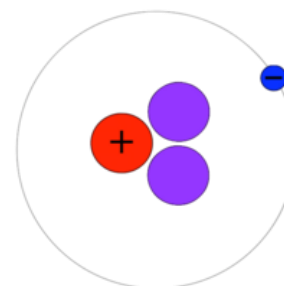
Protium



Deuterium



Tritium



Atomic Weight

1.0079 u

2.0141 u

3.0160 u

Half-Year

Stable

Stable

12.32 yr

Abundance

99.9855%

0.0115%

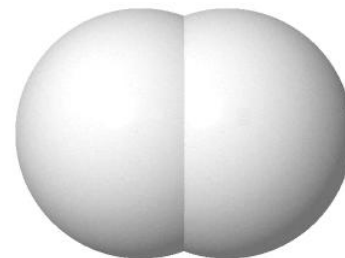
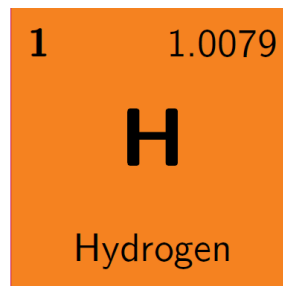
trace

Overview of Hydrogen

□ Hydrogen

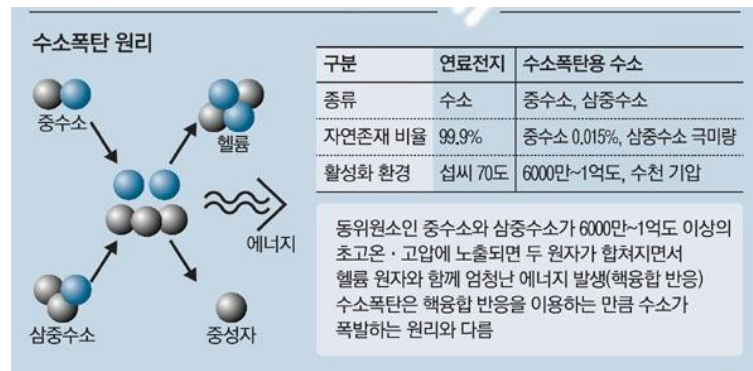
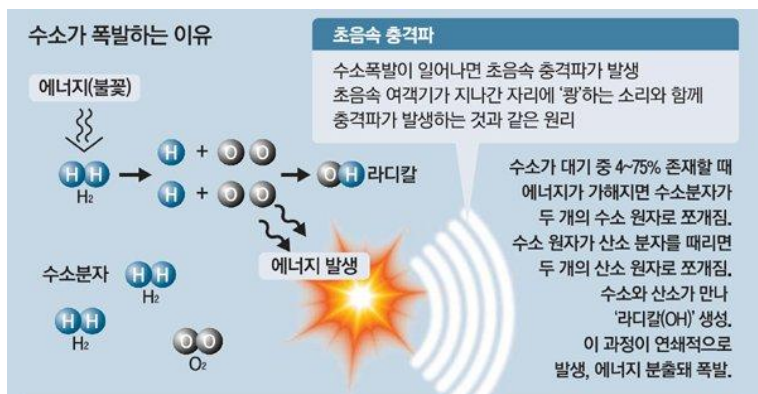
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–Low reactivity at room temperature, but reacts with most elements and compounds at high temperatures or in the presence of catalysts

- **Wide Flammability Range:** 4–75% in air (cf. 5–15% CH_4 in air)
- **Low Ignition Energy:** Minimum ignition energy of $\sim 0.02\text{ mJ}$, corresponding to $\sim 7\text{--}10\%$ of that required for gaseous CH_4 or gasoline vapors

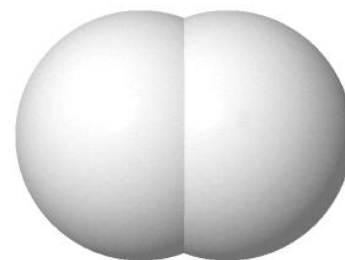
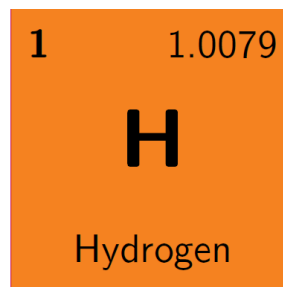


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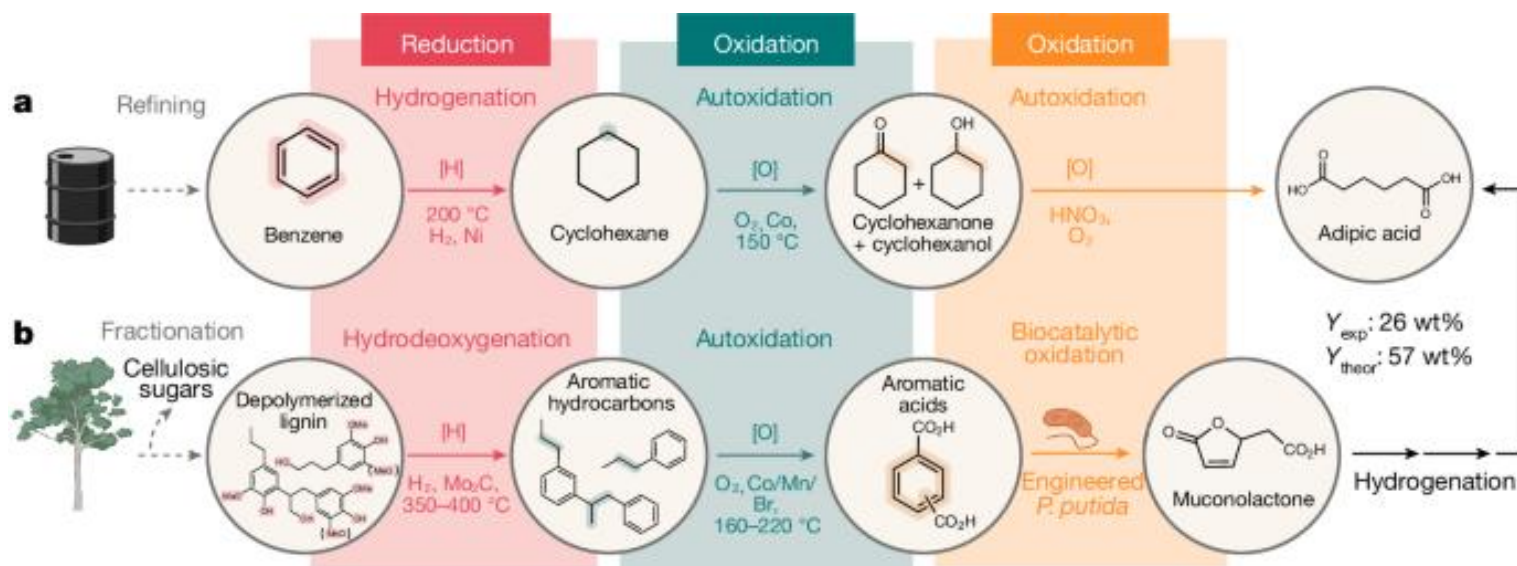
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–Involved in biological redox (oxidation–reduction) reactions

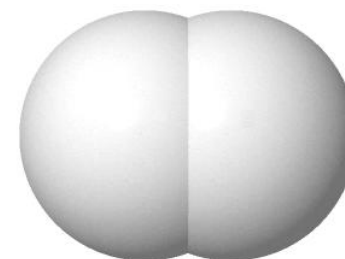
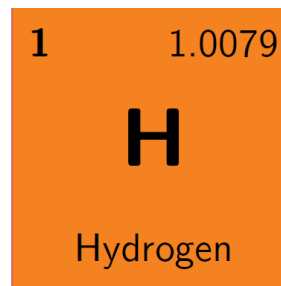


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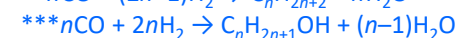
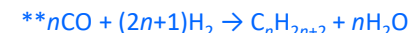
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–Reacts with halogens to form hydrogen halides, and with CO to produce hydrocarbons and alcohols



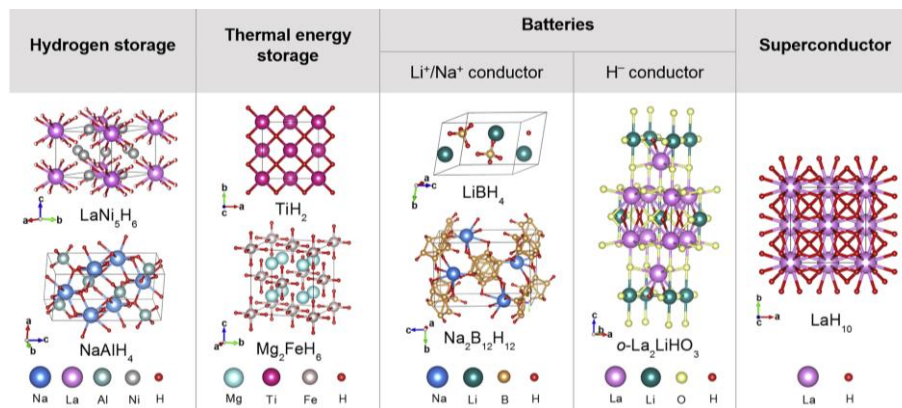
–Combines with numerous elements to form various hydrides

*Ionic hydride (groups 1, 2 except for Be, Mg) (e.g., LiH, NaH, CaH₂, etc.)

**Covalent hydride (groups 13–17) (e.g., B₂H₆, CH₄, NH₃, H₂O, HF, etc.)

***Metallic hydride (typically transition metals) (e.g., TiH₂, PdH_x, etc.)

***Complex hydride (e.g., NaBH₄, LiAlH₄, NaAlH₄, etc.)

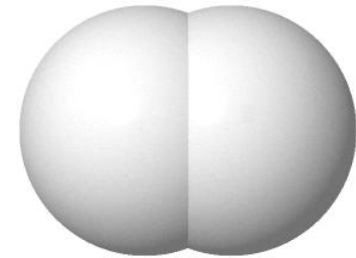
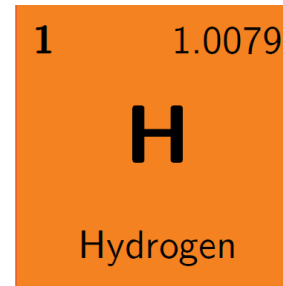


Overview of Hydrogen

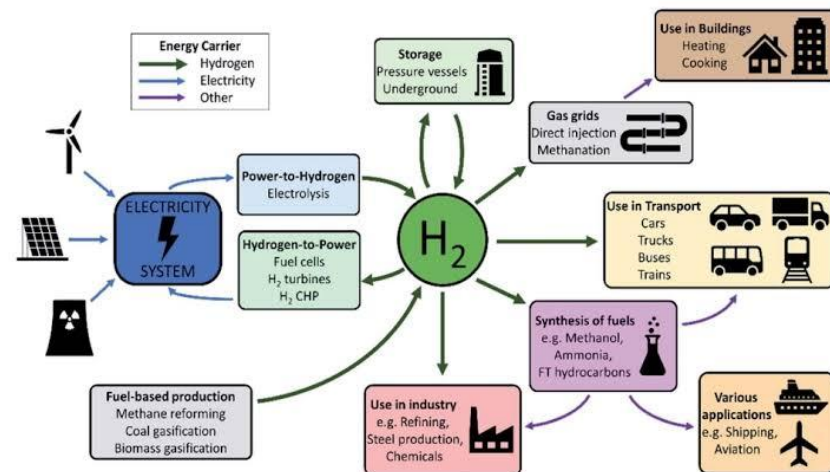
□ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: $-259.2\text{ }^{\circ}\text{C}$
- Boiling Point: $-252.9\text{ }^{\circ}\text{C}$
- Density: 0.090 g L^{-1} (at STP ($0\text{ }^{\circ}\text{C}$, 1 atm))



– H_2 is an *energy carrier* (NOT an energy source!) that is converted from primary energy sources into a secondary form, making it **storable** and **transportable**.

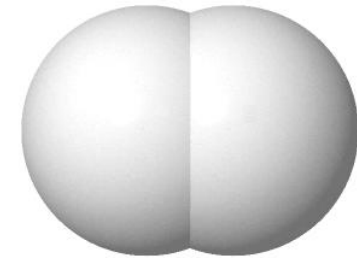
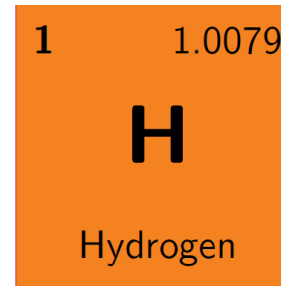


Overview of Hydrogen

□ Hydrogen

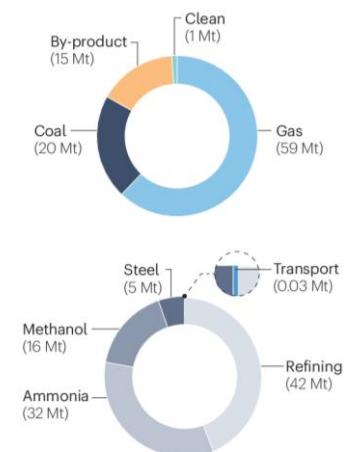
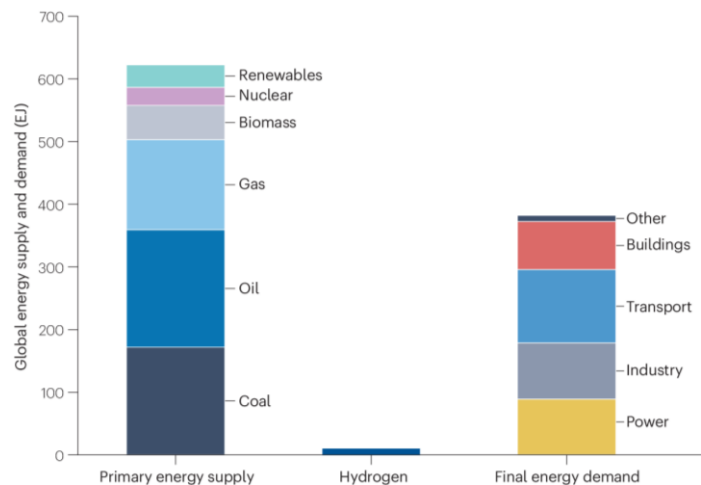
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–Major challenges include production costs, storage limitations, the development of highly efficient combustion engines, and safety; **the economic viability needs be improved**

The Hydrogen Challenge



Quiz



❑ Why Is Hydrogen So Scarce in Earth's Atmosphere?

H_2 makes up only about 0.55 ppm of Earth's atmosphere (See Slide #11). Even though Earth is known as the "water planet," atmospheric H_2 is extremely scarce. A common explanation often states: *"Because H_2 is the lightest molecule, it continually rises through the atmosphere due to buoyancy and eventually escapes into outer space."*

Is this explanation physically accurate? If not, what is the correct scientific mechanism behind the low concentration and escape of H_2 from Earth's atmosphere?

(Hint 1)

A helium balloon floats because surrounding air pushes it up. But what happens to a balloon when there is no air left to push it? Can "buoyancy" kick something entirely out into space?

(Hint 2)

Below 100 km, the Earth's atmosphere is constantly stirred by strong winds and turbulence. Does H_2 really rise to the top by itself?

(Hint 3)

Consider the Maxwell–Boltzmann distribution for root-mean-square speed ($v_{\text{rms}} = \sqrt{3k_b T/m}$)^{0.5} vs. Earth's escape velocity $v_{\text{esc}} = 11.2 \text{ km s}^{-1}$ (k_b : Boltzmann constant, T : Temperature, m : mass)

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